

Problem Set 6

Instructor: Robert Rand

1 The Chinese Remainder Theorem

Problem 1: Use the Chinese remainder theorem to solve for x , where

$$x \cong_3 2$$

$$x \cong_4 1$$

$$x \cong_5 3$$

2 Wilson's Theorem

Problem 2: Solve for the following:

$$\frac{103!}{101} \bmod 101$$

Don't use a computer and show your work.

3 Fermat's Little Theorem

Problem 3: Let p and q be distinct primes that are greater than $a > 0$. Show that

$$a^p \bmod p = a^q \bmod q$$